ARTICLE IN PRESS

Egyptian Informatics Journal xxx (xxxx) xxx

Contents lists available at ScienceDirect

Egyptian Informatics Journal

journal homepage: www.sciencedirect.com

Review

Security and privacy issues in e-health cloud-based system: A comprehensive content analysis

Nureni Ayofe Azeez *, Charles Van der Vyver

School of Computer Science and Information Systems, Faculty of Natural and Agricultural Sciences, Vaal Triangle Campus, North-West University, South Africa.

ARTICLE INFO

Article history:
Received 26 August 2017
Revised 11 October 2018
Accepted 17 December 2018
Available online xxxx

Keywords:
E-Health
Security and privacy
Cloud
Vulnerability
Access control

ABSTRACT

The recent advancement in Information and Communication Technology (ICT) has undoubtedly improved services in all sectors in the world. Specifically, Information Technology (IT) has led to a very vital innovation in health sector called electronic health (e-Health). In order to optimize full and excellent benefits of this innovation, its implementation in a cloud-based environment is important. However, with noticeable and numerous benefits inherent from e-Health in a cloud computing, its full utilization is still being hampered by challenges of security and privacy. In this paper, we focused on extensive review of current and existing literatures of various approaches and mechanisms being used to handle security and privacy related matters in e-Health. Strengths and weaknesses of some of these approaches were enunciated. The literature review was carried out after selecting over One Hundred and Ten (1 1 0) original articles and figured out several models adopted in their solutions. After comparing models used, we arrived at the reviewed articles. Reviewed articles were narrowed down to the current number because of similarity observed in the models adopted by some researchers. Also, we give an acceptable and standard definition of e-Health. Effort was made to classify cloud-based models. Security and privacy requirements as recommended by Health Insurance Portability and Accountability Act (HIPAA) were also discussed and provided. Remarks and recommendations were made regarding the review process and future directions on security and privacy of e-Health in cloud computing was also provided. Finally, authors propose a secured and dependable architecture for electronic health that could guarantee efficiency, reliability and regulated access framework to health information. The architecture, though is currently under implementation, will guarantee absolute security and privacy between healthcare providers and the patients.
© 2018 Production and hosting by Elsevier B.V. on behalf of Faculty of Computers and Information, Cairo University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Contents

1. Introduction . . . 00
2. Methodology. . . . 00
   2.1. Research questions . . . 00
   2.2. What is e-Health? . . . 00
   2.3. Cloud based e-Health models . . . 00
      2.3.1. Private cloud . . . 00
      2.3.2. Public cloud . . . 00

* Corresponding author at: School of Computer Science and Information Systems, Faculty of Natural and Agricultural Sciences, Vaal Triangle Campus, North-West University, South Africa. Tel.: +2347066838551.

E-mail addresses: nurayhn1@gmail.com (N.A. Azeez), Charles.VanDerVyver@nwu.ac.za (C.V. der Vyver).

Peer review under responsibility of Faculty of Computers and Information, Cairo University.

Production and hosting by Elsevier

https://doi.org/10.1016/j.eij.2018.12.001
1110-8665/© 2018 Production and hosting by Elsevier B.V. on behalf of Faculty of Computers and Information, Cairo University.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

2.3.3. Hybrid cloud . . . 00
2.4. Benefits and demerits of e-Health . . . 00
2.5. Privacy and security requirements in e-Health . . . 00
3. Literature review . . . 00
4. Remarks and recommendations . . . 00
5. Future directions on security of e-Health . . . 00
6. Features of proposed system . . . 00
7. Conclusion . . . 00
References . . . 00

1. Introduction

The evolution of Information and Communication Technology has influenced the traditional approach for healthcare practices in the world. This development is well noticed in a partial abandonment of paper-based medical prescription to electronic version particularly in most of the developed countries of the world [1]. The need to federate and integrate various electronic health information from various domains such as medical research laboratories, hospital, health insurance firms has led to the evolvement of a concept called electronic health (e-Health). Simply put, e-Health can simply be defined as the use of Information Technology (IT) infrastructure and e-commerce practices for processing, sharing and manipulation of health information.

It is however noted that different domains being involved in sharing of medical data have made the application very difficult to manage hence the need for cloud-based environment which allows collaborative sharing of information across multiple administrative domains [1]. Cloud computing has so many advantages among which is seamless transfer and sharing of medical information in a timely manner. It has also relieved healthcare providers the rigour involved to manage infrastructure and also provide them ample opportunity to familiarize with IT service providers [2]. It has been established in different academic papers that cloud computing offers numerous benefits ranging from scalability, cost effectiveness, agility enhancement of collaborative sharing of resources [3].

Despite its various advantages, there are security and privacy challenges that urgently deserve utmost attention for realization of its efficient and full scale utilization [4]. Cryptographic and non-cryptographic approaches have been used on several occasions to ensure preservation of security and privacy of health data in the cloud computing. Also, fine-grained as well as patient-centric access control schemes are commonly being used to achieve privacy in electronic health. In this paper, various security measures being used for protecting data are reviewed. Their strengths and weaknesses are also exposed. Effort was made to proffer better alternatives for securing e-Health data.

2. Methodology

It is practically impossible to review all the available literatures that are related to security and privacy in e-Health. Consequently, we succeeded in reviewing selected articles. The literature review was carried out after selecting over One Hundred and Ten (1 1 0) original articles and figured out several models adopted in their solutions. In order to have a reasonable and appreciable number of reviewed articles, we downloaded over forty (40) articles from ACM digital library, fifty-seven (57) articles from IEEE digital library and forty-three (43) from IEE *Explore* digital library. Other papers were downloaded from Springer, Elsevier and Science Direct digital libraries. Few papers were downloaded from other journals that are not popular and rated as those mentioned. After comparing models and approaches used by many researchers, we arrived at 110 reviewed articles. Reviewed articles were narrowed down to the current number because of similarity observed in the models adopted by some researchers.

What is more? We specifically reviewed the articles by examining and analyzing strengths of each of the methods adopted in finding solution to the security challenge in e-Health. We also identified various weaknesses of each of these methods and finally proposed the way forward for combating security and privacy lapses in e-Health.

2.1. Research questions

In order to carry out this research work, the following research questions were posed:

i. how do we identify security and privacy lapses in e-health?
ii. how can identified challenges identified in 'i' be solved in order to guarantee security of e-health?
iii. what are the future directions on security and privacy of e-health?

2.2. What is e-Health?

E-health is an emerging field in the intersection of medical informatics, public health and business, referring to health services and information delivered or enhanced through the Internet and related technologies. In a broader sense, the term characterizes not only a technical development, but also a state-of-mind, a way of thinking, an attitude, and a commitment for networked, global thinking, to improve health care locally, regionally, and worldwide by using Information and Communication Technology [5].

2.3. Cloud based e-Health models

Three types of cloud models are usually employed for rendering e-Health cloud services. They are: private, public, and hybrid cloud.

2.3.1. Private cloud

This model is considered the most secured of all the models. There is a complete restriction to the public internet. The Electronic Medical Records (EMRs) in a private cloud are only accessed by a recognised personnel of the healthcare institutions who are regarded to be trustworthy and reliable [6]. A private e-Health cloud is shown in Fig. 1.

2.3.2. Public cloud

This model consists of shared infrastructure that is in total control of the third-party provider. The services of this form of cloud system are procured from the Cloud Service Providers (CSPs). With this model, EHRs are usually shared among various organisations. The EHRs are very vulnerable to various attacks and manipulations because they are stored at off-premises servers that are under the control of CSPs. To circumvent this security challenge, efficient cryptographic mechanisms and fine-grained access control

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

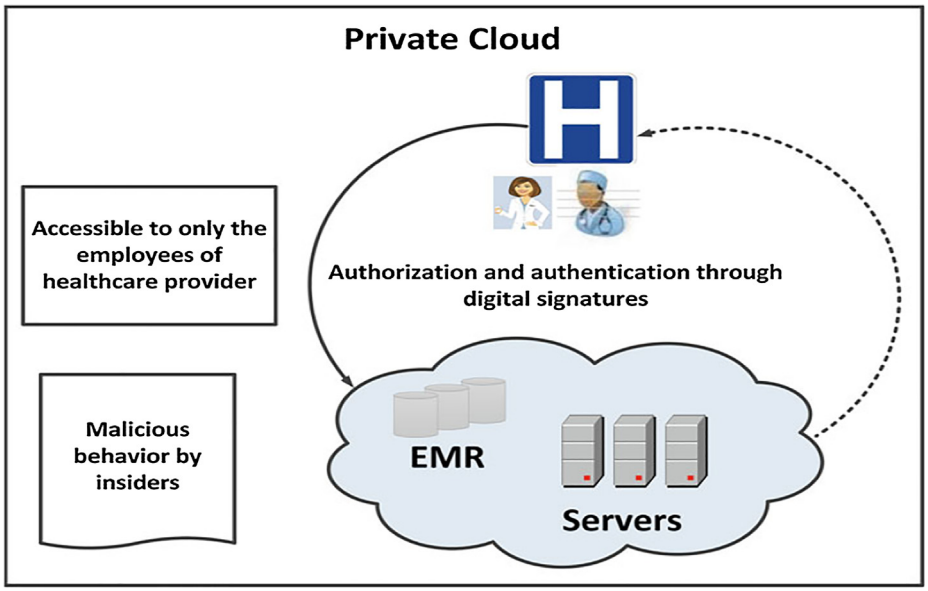

Fig. 1. An example of a private cloud in context of e-Health [7].

frameworks are required [6]. An example of public cloud is depicted in Fig. 2.

2.3.3. Hybrid cloud

This is a combination of both the private and public clouds such that each of the models operates individually but united through standard technologies [8]. The deployment of this model for e-Health is highly advantageous since it combines benefits of both models (public or private). Healthcare providers with restricted and limited physical resources as well as strong interest in using legacy systems can conveniently make use of third party services to house big medical data [9]. It however requires efficient security framework before it can be maximally utilised. An example of hybrid cloud is depicted in Fig. 3.

2.4. Benefits and demerits of e-Health

As presented by [10], a typical e-Health application should be able to provide the following services: Findings for emergency e-Health services, multimedia conferencing, streaming of medical images, Tele-Robotic Systems, transmission of patient vital signs, findings for emergency e-Health services, access to Electronic Health Records and research & education.

2.5. Privacy and security requirements in e-Health

Achieving security and privacy in e-Health is very vital in achieving the objectives of using this modern technology [46]. This is very necessary as digitizing health related data and sharing them

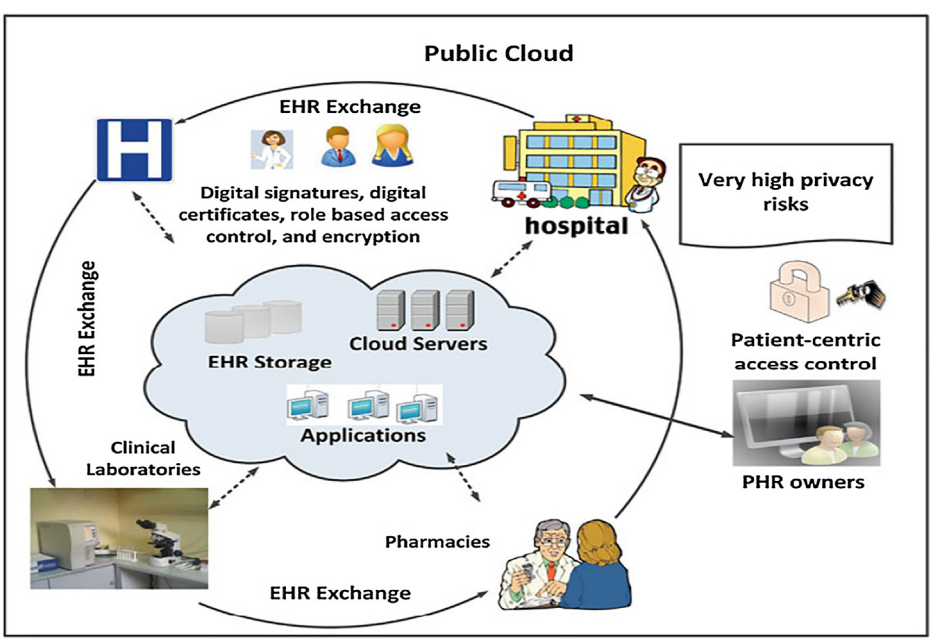

Fig. 2. An example of a public cloud in context of e-Health [7].

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

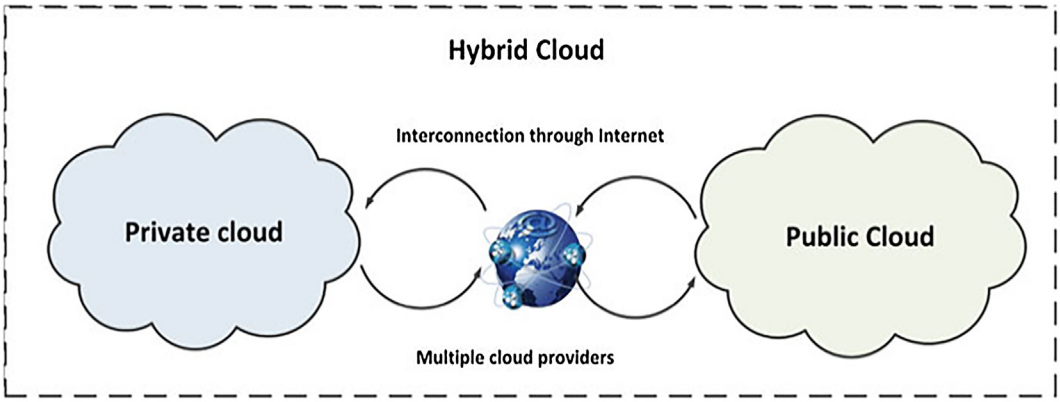

Fig. 3. An example of a hybrid cloud in context of e-Health [7].

may lead to different forms of attacks [11]. Many government health institutions have therefore developed framework to ensure high level of security and privacy. For instance, the Health Insurance Portability and Accountability Act (HIPAA) was put forward by the United States (US) Congress in 1996 as a federal law that applies for the US healthcare industry. In accordance with HIPAA [11] guidelines, a set of valuable security and privacy requirements must be put in place for effective utilization of e-Health. These are presented in Table 1.

3. Literature review

This section presents review of several articles in journals, conference proceedings, documents from the internet, book chapters and books on various security approaches and mechanisms being used in e-Health. We did identify benefits and demerits of each of the approaches. We also provided, in the next section, how those weaknesses could be taken care of.

Shin et al. (2014) examined various security models for healthcare applications and attempted to see how information leakage could be protected. They evaluated various security requirements to ensure security and privacy in electronic health. To find solution to identified security challenges in electronic health, they employed extended Role Based Access Control (RBAC) security model. They came up with u-healthcare service integration platform where extended RBAC model was deployed. The architecture was designed to carry out four main functions: exchanging health information, meal recommendation, transaction of health information and management of health information on any smart devices [12]. It is however worthy of note that security issue was not properly resolved. The model is not suitable for any distributed environment. As a result, the solution provided has limited applications. The application does not also consider expansion in the number of users.

Simplicio et al. (2015) demonstrated how a lightweight framework was used to present SecureHealth architecture that is based on Transport Layer Security/Secure Sockets Layer (TLS/SSL) for protecting data exchange with server that requires no extra security layer [50]. SecureHealth which includes many security features like authorisation provides security services for transmitted and stored data. It has a good benefit of preventing alien from unauthorized access to the system that contains health information. Aside from this, it provides the manager the capability of identifying misnomer from information supplied [13]. Despite the benefits accrue from this framework, the main challenge is that it is platform dependent and not scalable. In a cloud based environment, the security policy and framework must give room for scalability and future expansion.

In order to ensure that e-Health care service providers decrease the cost of maintaining data and allowing it to be available online in a secured manner, Barua et. al., proposed a security mechanism with different level of hierarchy. Provision of access control was carried out at a central level. They adopted Attribute Based Encryption (ABE) in such a way that privileges were mapped and juxtaposed into various roles with ABE access structures. The main challenge with this approach is the complexity of responding to various requests from different users due to storage of health information located in a centralized server [14]. Also, priority needs to be set when there is a simultaneous request by users.

In order to solve the challenge of having data storage of health information in a centralized server, Guo et al. (2012) considered the distributed and collaborative nature of e-Health system. They didn't allow a centralized server to handle authentication and authorization procedures, instead, they allowed both the patients

Table 1
Security and privacy requirements as recommended by HIPAA.

Requirement	Description
Patient's understanding	This implies that patients have an exclusive right to know and understand how their sensitive and private health information are kept and utilized by any healthcare provider.
Patient's control	This allows patients to be given permission to determine who can access his/her health data.
Confidentiality	Health information should be kept away from people who should not access it. The sanctity of the information should be maintained.
Data integrity	This ensures that manipulation and omission of health information is totally prohibited. Hence, health information being shared should be a true representation of original information without any form of amendment or alteration.
Consent exception	This stipulates that patient's information could be accessed without his consent only in emergency cases.
Non-repudiation	Healthcare practitioner should deny the fact that it has performed a certain activity on the sensitive data of patient. Such activity should be supported with evidence to avoid dispute or suspicion.
Auditing	This is a requirement that health data should be well monitored frequently along with any form of activity to ensure that data is well secured and protected. This will assist user to know the confidential status of his data.

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

and doctors to carry out authorisation process. In fact, users are given access based on their privileges without disclosing their attributes and identities. This framework addresses and solves the problem of handling and maintaining security, privacy as well as variability of all users' attributes [15]. However, there is no room for collaborative sharing of medical data across different domains. The framework is too complex to implement. As of now, there is no real-life implementation to proof its efficiency as claimed by the authors.

Gajanayake et al. (2016) came up with a special privacy and secured oriented access control architecture only for e-Health. They achieved their design by combining three different security models: Discretionary Access Control (DAC), Mandatory Access Control (MAC) and Role Based Access Control (RBAC) [16] to come up with a novel architecture which allows healthcare providers and patients to dictate and set access privileges. The major drawback of this framework is in its ability to be useful only as a standalone security model to attain health electronic record requirements [17].

A secured patient-centric electronic health information schema was proposed by Barua et al. for providing reliable access privilege in a cloud-based environment by using a protocol called Proxy Reencryption. The schema which has five main stages makes use of Attribute Based Encryption to permit patient-centric access control. The performance analysis reveals that the schema has a good and excellent performance. The weakness is that it is not flexible enough for other form of distributed systems. What is more, the schema doesn't give room for scalability and flexibility [14]. Only limited number of users was considered during evaluation.

Kumar et al. (2013) proposed a new framework for electronic health on encryption technique; Attribute Based Encryption (ABE). In this case, users are divided into two principal domains: personal and public domains. The essence of this is to handle key management complexity [45]. In the personal domain, every owner is allowed to encrypt/access only data under his attributes while public domain allows users to adopt and make use of multi-authority ABE to enhance the security countermeasures [18]. The great challenge with this approach is the issue of scalability and flexibility because integrating Attribute Based Encryption into large scale Electronic Health Record system poses serious and great key management challenge.

Zhu et al. (2014) proposed a secure and reliable framework that makes use of re-encryption and Attribute Based Encryption (ABE) with proxy encryption that is Rivest Shamir and Adleman (RSA) enabled. The objective of using proxy was to introduce separation mechanism to guarantee the validity of patient's data. In this case, only the professionals are given the write privilege keys while the read privilege keys are given to patients [19]. The essence of this is to prevent full authorization by the patients. Through this framework, the computation overhead has been drastically reduced. With this approach, the healthcare practitioner can easily be prevented from getting the read keys without approval from both end. However, the scheme gives room for limited number of users.

Another secured model was proposed by Sunagar and Biradar. The secured framework is based on Advanced Encryption Standard (AES) algorithm developed to encrypt information about patients based on the security policy [20]. The security model allows users to maintain information in a reliable and secured manner in a cloud-based environment. The framework which has three modules, guarantees high level of security and privacy. The drawback of this framework is that it could not work with all forms of operating systems. It is operating system dependent. It is also very complex to implement in a real life scenario.

Liu et al. (2015) combined Role-Based Access Control (RBAC) with Hierarchical Identity-Based Encryption (HIBE) schema to come up with an encryption technique for securing patient's data before they are outsourced to the storage data [21]. RBAC assists to facilitate user's privileges. The main weakness of this framework is that it doesn't provide reliable and accurate access control requirements. Patient may not access privileges to their confidential information without following HIPAA regulations.

Bahtiyar and Cağlayan proposed from the view point of an entity, a trust-based assessment security model for electronic health services [22]. The model which contains a detailed architecture that is applicable to various entities and serve as a unique trust assessment metrics may be used to evaluate a specific feature of a security system. The results obtained from simulation revealed that the framework gives better results in terms of trust computation when compared to the various existing trust models for e-health solutions. The model is however cumbersome, it contains some mathematical variables whose values were not clearly evaluated.

Shin et al. (2014) evaluated various security requirements related to privacy in electronic healthcare services. They proposed an enhanced Role Based Access Control model to design u-healthcare service integration platform (u-HCSIP). The model which carries out four main functions is however unusable in a collaborative environment. It doesn't permit user to have exclusive privilege to dictate who should access his/her medical details [12]. The model is not implementable in a cloud-based environment.

The work of Li and Hoang centres on a novel role-interaction-organisation privacy and security model with specific application to electronic health system and services [23]. The framework is modelled in form of a multi-agent system. The role in the model determines both access privileges and initiates various requests to interact in a flexible manner with agents who satisfy the security requirements. To confirm the efficiency of the model, a simple case from electronic health system was used for performance evaluation. The main drawback with this model is that it lacks security information aggregation.

The work of Fan et al. (2014) also underscores the need for privacy in e-Health system. They carried out a design and implementation with the Single Point of Contact (SPoC) which guarantees claim-based authorization and as well facilitates integration and deployment of reliable e-Health services to be hosted in a cloud-based domain [24]. The result of the model is fairly reliable. The application can only work with limited number of users. It is not flexible and dynamic enough for a very large number of users.

Bhartiya et al. (2015) work on access control security model that used a unique Hierarchy Similarity Analyzer (HSA). The model evaluates and assigns a Security_Level (SL) to users sharing data across different administrative domains. The SL ensures the approved and authorised percentage of data to be shared on any agreed collaboration of different policies. This security model allows combination of various policies and recognises the likely policy-disparity culminating because of attribute conflicts in the set of defined rules [25]. The model was implemented using XACML policies and compared with other similar security models. One of the major challenges with this framework is in its inability to incorporate different types of policy-conflicts such as temporal constraints, semantic and syntactic. Also, there is no absolute guarantee to security and privacy in any federated agencies.

Rezaeibagha and Mu (2016) developed a novel access-control framework to address the challenges of security and privacy in Electronic Health Record (EHR). The framework adopted hybrid clouds as well as access control policy transformation to guarantee reliable and dependable access control and authorization-preserving data sharing among different healthcare providers [26]. To make the model efficient, some cryptographic building blocks were introduced with access control policy transformation to tackle different users of EHR with different access privileges and permission in various cloud environments. The main drawback

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

with this framework is its inability to give room for user's expansion. It doesn't give room for scalability because the number of users is limited.

In an attempt to find a solution to security issues in EHR, the duo of Garcia-Morchon and Wehrle developed a fine-grained access control for ubiquitous healthcare electronic applications. The framework enhances the existing traditional RBAC security model for two purposes. It explores it to allocate access policies to different sensor nodes and also stores very vital information such as health information, time and location that are very critical to decision making on security [27]. The modular nature of the framework makes it easy and convenient for deployment of policies on various sensor networks. One of the main challenges of the model is lack of detection mechanism for illegal and unauthorised access in case of any emergency situation.

Efforts were made by Amini et al. (2011) to come up with a lightweight security model for e-Health. In achieving this, they look into various set of security protocols like MiniSec, RC4-based as well as various ciphers algorithms such as RC4 and Advanced Encryption Standard (AES). The researchers applied the ciphers algorithms to a combined attacks [28]. At the end of the experimentation, they confirmed that Skipjack cipher algorithms and RC4 are very efficient and reliable for achieving access confidentiality and integrity in Electronic Health. The authors failed however to investigate the efficient approaches for the remaining security requirements. Invariably, the conclusion drawn on the few algorithms investigated could not justify the conclusion that they are the best and most efficient.

Since it has been established that traditional Public Key Infrastructure (PKI) for implementing cryptographic mechanisms is cumbersome and time consuming, Wang et al. (2017) described different related cryptographic techniques for ensuring security and privacy of Electronic Health system. They evaluated the performance analysis of these techniques which include Identity Based Encryption (IBE) and new Identity Based Proxy Re-encryption (IBPRE) schemes [29]. From the evaluation, it was observed that newly developed IBPRE is better and efficient for re-encryption which can subsequently be used to protect health information in the cloud. The drawback with this technique is that, the authors were unable to verify the performance of other encryption techniques hence should not be categorical on the efficiency of the new IBPRE.

In a bid to secure medical data and other sensitive medical information, Karakış et al. (2015) achieved this through the combination of medical data into one single file by using steganographic approach for hiding data. In their work, they proposed two new image steganography approaches that are dependent on fuzzy-logic and similarity. The objective of this was to allow for selection of the non-sequential least significant bits (LSB) of image pixels. They made use of the similarity values of the realised gray levels in the pixels hide the message [30]. With this approach, the message is protected to ward off any form of attack through lossless compression and symmetric encryption algorithms. The performance of steganographic image quality and rating was measure by Mean Square of Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Measure (SSIM), Universal Quality Index (UQI), and Correlation Coefficient (CC). With the results obtained, the newly proposed approach guarantees security and privacy of patient information and also increases data repository [49]. The drawback of this approach is that it could not handle and tackle noise cancellation and data reduction which might enhance embedding capability.

After acknowledging the vulnerability nature the cryptographic-based approaches for securing health records, Sahi et al. (2016) did some technical review on some other security models being used to secure and protect electronic health information. They eventually proposed two main approaches to ensure security and privacy. The approaches are the Privacy-Preserving approach and the Security-Preserving approach on one hand and a disaster recovery plan on the other hand [31]. The former approach is a robust mechanism for achieving both the privacy and integrity of medical information while the latter approach could only be used for reliable and dependable authentication approach for electronic medical information. The main drawback is the inability of the mechanism to function efficiently if any of the approaches fails. The model is not also interoperable enough to be accommodated and implemented in a cloud-based environment.

Peleg et al. (2008) critically studied Role Based Access Control (RBAC) and observed that it didn't provide account for the reason under which request to access sensitive data is made. As a result of this weakness, they developed a framework with a Situation-Based Access Control (SitBAC) model. SitBAC is considered to be a conceptual framework that defines scenarios where a patient's access to electronic information is either denied or allowed [32]. SitBAC uses the Situation Schema which comprises of Patient's entities, Electronic Health Record (EHR), Data-Requestor along with their relations and properties. This model is considered to be generic which can also be used and adapted in other domains apart from medical information. One of the major weaknesses of this model is that it could not be able to factor in all the likely stakeholders with their various goals. Also, the model does not include a formal representation of the SitBAC as a knowledge base.

Rubio et al. (2013) modified the security features in the Standard communications protocol for computer assisted electrocardiography (SCP-ECG) to allow file to be stored securely. The new security approach allows SCP-ECG files to be properly accessed (permitted or denied) to users for numerous reasons ranging from clinical teaching or research, interpretation of the test as well as consultation. Access privileges are supported by cryptographic elements that are well induced and scaled by means of role-based profiles [33]. The application has been confirmed to be very effective to authorise and authenticate users and protect the privacy of sensitive electronic health information [47]. Despite the efficiency of the framework, the application cannot be deployed in a cloud based environment hence there is difficulty in using the model in a distributed environment. The model is technical and difficult to implement. There is no real-life implementation to justify its efficiency and capability.

Martínez et al. (2013), considered anonymization as a means of securing electronic health information. They opined that the use of social security number wasn't sufficient for anonymization as there is tendency for disclosure of information that are very vital to patient. Against this backdrop, they proposed Statistical Disclosure Control (SDC) approaches to protect and mask very confidential attributes while preserving to a reasonable level, the features and utility of anonymised data. These approaches focus on continuous-scale numerical data only as against the common non-numerical attributes which their application to EHR is far from optimal results. The framework proposed by Martínez et. al. provide accurate and reliable results of SDC approaches to non-numerical clinical data with special attention on the preservation of semantics [34]. To achieve their objective, they used structural knowledge bases such as SNOMED CT. This framework is however applicable only to non-numerical attributes. This framework cannot also work for large dataset. Also, the framework generates a significant delay during the anonymization process

Some of the anonymization approaches already proposed by researchers are characterized with significant delay during the anonymization process because they focused majorly on applying existing security models to data obtained from data stream. It on this note that Kim et al. (2014), developed delay-free anonymization

for preserving and securing the privacy of health-based information. This approach didn't generate time delay since stream of data are anonymized immediately with fake values [35]. Also, late validation is also encouraged to enhance data utility of the anonymization results and at the same time managing the fake values. This method however did not consider the statistical analysis and old data considered for obtaining the values hence it was unable to obtain more dependable anonymization results. Also, it didn't investigate the timing for the late validation, as a result, the results obtained are not dependable.

Barua et al. (2011) proposed a secure framework named Efficient and Secure Patient-centric Access Control (ESPAC) which permits information requester on electronic health platform to have access privileges based on their assigned roles and their corresponding attributes. ESPAC moves data to a centralized storage thereby reducing the overall maintenance cost of moving data from one storage to another [36]. With this framework, Electronic Health Information is accessible anytime and anywhere [48]. The scheme also ensures that user privacy and data integrity are properly taken care of. The weakness of this scheme is that it didn't take into consideration other forms of Role Based Access Control (RBAC). It only considered core RBAC in handling it security and privacy issues.

Kester et al. (2015) observed that privacy and security of health-based information was important. To achieve this, they considered reversing encrypted and watermarked images so that plain images worked on in the processing of watermarking and encryption can be totally recoverable due to the sensitive nature of information conveyed in medical images. They proposed a recoverable watermarked and encrypted image processing technique for security and privacy of medical images [37]. The scheme is being used to secure and authenticate medical images. The framework is limited to images alone with factoring text and audio data.

Wu et al. (2012) also observed the need for having electronic health information in a cloud based environment to ensure interoperability and achieve lower operational cost. Their concern was majorly however on how to ensure security and privacy of the data stored in a cloud environment. In achieving a secure and dependable cloud environment where electronic health records (EHR) is stored, they proposed a systematic access control framework to aid selective sharing of composite and complex EHR combined from different healthcare providers in any cloud-based environment [38]. The framework has however not been tested with a real-life healthcare dataset to truly confirm its suitability and reliability. Also, the composition issue was not addressed as well as how to handle fine-grained delegation mechanism for data sharing in a cloud based environments.

Yu et al. (2010), attempted to solve the security challenges in Electronic Health by studying weaknesses known with cryptographic methods which is the introduction of a heavy computation overhead and data management problems whenever fine-grained data access is proposed. They solved these problems in two folds: first, by enforcing and defining access policies based on attributes of data. Second, by allowing data users and owners to delegate majority of computational task involve a modularized data access control to unreliable cloud servers without disclosing the contents of data [39]. This process was achieved by combining lazy re-encryption, proxy re-encryption and attribute-based encryption (ABE). However, this scheme has not proved to be fully scalable as also did not provide efficient and reliable data confidentiality and integrity.

Löhr et al. (2010) developed a security and access control architecture for establishing reliable and privacy domains in electronic health infrastructural resources. The architectural model provides a combination of users and network security. The framework has several unresolved challenges [40]. There might be a problem with patient to authenticate and access his e-Health information. This might be due to his inability to remember a PIN to access his information. If this arises, the problem of confidentiality of existence, client anonymity and challenge may arise.

Hupperich et al. (2012) developed a privacy and access control architecture for EHR that is more flexible but allows encryption of patients' data. In this platform, patients are able to authorize and allow access to their medical information from a far distance via a mobile device. The access is time independent and not place-bound. This framework relies solely on modern cryptographic security approach [41]. The main disadvantage of this scheme is in the time required in exchanging key from one end to the other for both encryption and decryption. The complexity of key management is a serious challenge of this approach.

Adoption of two stage-key access control and zero-knowledge protocol was considered by [42] for e-Health system. In order to obtain a secure connections between various entities DUKPT and a two-stage combination of key encryption was adopted. The framework was analysed with respect to resistance to common attacks and data confidentiality. The proposed scheme tolerates a good number of simultaneous authorization requests with excellent response time. The main drawback with this scheme is that it has limited number of entities: users U, a cloud server CS, a service provider SP, and an authentication and access control manager AAM. This implies that it is not scalable; this is not good for a cloud-based environment because it will not give room for collaborative sharing of resources.

Another framework for access control to Personal Health Records (PHRs) in a distributed environment was proposed by [43]. They leveraged Attribute Based Encryption (ABE) security model to encrypt patients' data in order to attain and achieve scalable access control as well as fine-grained access for PHRs. The system was divided into multiple security domains in order to reduce the complexity that may arise due to key distribution. In this case, each domain manages only a small percentage of the users. Users are given absolute control over his own privacy. The scheme is dynamic in nature as it supports on-demand withdrawal of user's access privileges. The scheme however fails to support more expressive owner-defined access control policies.

A novel patient-centric scheme with good framework for data access control to electronic health records was proposed by [4]. They identified scalability and fine-grained access as part of the challenges. They therefore optimized and leveraged Attribute Based Encryption (ABE) models to secure and encrypt patient's PHR data. They focused on a situation where there is multiple data ownership as well as a condition where users are divided into various security domains. This was done to reduce the key management complexity for both the owners and users. With the exploitation of multi-authority ABE there is a better performance of patience privacy and security. The scheme allows flexible modification of various file attributes. Despite the claims by the authors, the proposed framework was only simulated and it has not been tested on a real life cases to establish the claim. The results obtained could the not guarantee efficiency of the framework when tested in a real-life situation. Also, few users were also considered. The authors failed to inform what may happen when there is increase in the number of entities.

FinTech is globally considered a unique innovation adopted by various financial organizations across the globe. This technological advancement in financial sector covers a broad range of innovation such as service deliveries and data security. In a bid to create absolute and required awareness of FinTech technologies particularly for professionals and academia, Gai et al. (2018) did a comprehensive survey of FinTech through reviewing and collection of recent development and progress recorded. In their work; which is considered relevant and widely recognized, they summarized five

technical areas considered sacrosanct which are data techniques, applications and management, service models, hardware and infrastructure as well as security and privacy. They concluded their research by developing excellent and dynamic solutions for FinTech tagged "Data-Driven FinTech Framework (DF2)" to ensure upcoming FinTech procedural deployment [52].

With the current expansion in the development of networking solutions, various challenges have surfaced which are currently affecting effective and excellent usage of this brilliant innovation. The problem of security and privacy is considered a major challenge [44]. Basically, the process of achieving encryption is facing problems while data processing is not really applicable over ciphertexts. Having noticed this great challenge [53], proposed a technique for blending arithmetic processes and actions on a tensor based fully homomorphic encryption (FHE) which emphasizes calculation of encrypted real numbers. The main technique of their work was the adoption of DaC to increase a polynomial based mathematical operations and divide it into binomials. The results from their findings show efficiency and reliability of encryption-decryption technique which could satisfactorily get good outputs from decrypting cipher-results [53].

Due to the ubiquitous nature of mobile devices, specifically, iOS devices and Android as well as other Internet-based (connected) devices, the necessity and urgency to guarantee and ensure adequate and efficient security of data at-rest and data-in-transit without incurring performance expenses is very crucial. Having identified security and privacy as one of the major challenges for optimizing the great benefits of effective utilization of these devices, [54] proposed security and privacy model that is structured and designed to increase the entire security level with the aid of dynamic programming. An NP-hard defined problem attempts to utilize the value of complete protection weights defined and described by the security classifications. Gai et al. finally demonstrated the effectiveness of this model by using an Android app prototype to showcase its utility [54].

The booming development of the internet-based practices and techniques has undoubtedly enhanced the uprising of network based applications. It has been established that a successfully connected environment brings about the amalgamation of different techniques and strategies such as Internet-of-Things (IoT) and cloud computing. From available literatures, security and privacy matter has been the major challenge for data transmissions which is linked to poor privacy and protection communication protocols. In an attempt to find a lasting solution to this challenge [55], proposed a solution that focuses mainly on a unique strategy for implementing a high security transmission via the adoption of multiple-channel communications. They successfully evaluated the performance of the proposed approach through implementation and testing and confirmed that the approach could guarantee efficient and reliable level of privacy [55].

It is very clear that the positive and beneficial impacts as well as applications of medical decision support system with the adoption of data mining approaches cannot be over-emphasized. Apart from reducing consultation and diagnosis hours it could also assist in enhancing the diagnosis precision and accuracy. Naïve Bayes has been in use for quite some times to unearth useful information in order to enhance clinical and medical support system. With some of the accrued benefits of this innovation, its full-scale adoption and utilization has been hampered and vulnerable due to security and privacy concerns. It is in recognition of this great challenge that [56] developed a novel 'security-conserving patient-centric clinical decision solution which assist physician to diagnose the risk of patients in a very secured and reliable manner. In their work, the patients' legacy information are conserved in the cloud which can be used to train the algorithm (Naïve Bayes classifier) without revealing any vital information about patient. To finally achieve their objective, they designed an additive homomorphic proxy aggregation scheme which is a new cryptographic tool. Not only that, they introduced a privacy-preserving top-k disease names retrieval protocol to leverage the outflow of the Bayesian classifier. The approach does, however, not take into consideration some other data mining techniques [56].

Because of various advances in Information and Communication Technologies (ICT), it has been observed that clinical decision support systems is gaining ground and becoming highly impactful in the global health sector. Despite immense and recognized benefits from this great innovation, patients are becoming highly vulnerable due to the absence of adequate security and privacy measures to guarantee safety of information. Against this backdrop [57], proposed a novel security-preserving clinical decision support initiative called Peneus. This framework could be applied to monitoring health status as well as prediction of diseases. To enhance this framework, they went further to design a unique integer number calculation strategy over filled ciphertext in Single instruction, multiple data (SIMD) style. The work proves to be very efficient even though, construction of dependable bootstrapping methods was not included as well as a platform to care for large volume plaintext domain [57].

4. Remarks and recommendations

For any e-Health application to be very efficient and reliable, strong security mechanism must be put in place to effectively counter various threats such as tampering, denial of service attacks, spoofing identity as well as any form of unlawful and unauthorised privilege escalation. Also, the security mechanisms should be able to guarantee the privacy and protection needs that may be required as a result of the architecture of the cloud system. The Public Key Encryption (PKE) is widely being used for taking care of several security requirements such as anonymity requirements, collusion and unlinkability. However, with the extensive and prevalent adoption of PKE infrastructure, there is need to improve and enhance the capabilities and efficiencies of PKE. This is very important to guarantee integrity, confidentiality, reliability as well as authenticity of the health-related data in either a private, public or hybrid cloud.

Another important security scheme which has been extensively discussed is ABE. This security model satisfied some of the security and privacy requirements. ABE has also been found to provide fine-grained and excellent patient-centric access to electronic data [51]. With this approach however, ABE-enabled scheme is expensive when considering computational complexity. There is also a big challenge when it comes to management of access policies [49].

It has been established that there is a great increase in both size of ciphertext and key with the number of available attributes. Ciphertext in ABE is considered to be very expansive in usage hence decryption actions are very expensive in terms of computation. Because of this, the use of ABE has been limited in power and energy controlled handheld devices [44].

The approaches being used to achieve accountability, transparency and audits of health data need to be enhanced and supported. Adopting any of the three approaches mentioned will assist to provide security and privacy [48]. They will also assist in identifying how the authorised entities are making use of health-related data. The challenging issue here is how to identify and confirm if an entity is strictly adhering to the standard protocol in the normal way or not.

Revocation of access privileges given to any authorised entity on health data is another important issue that deserves attention. The specification of the access policies and privilege for different types of users from different domains is very challenging in any e-Health cloud environment. The key management should be

done in such a way that the need to redistribute key is totally eradicated [45].

What is more, selective encryption is also very important when carrying out security measure on e-Health data. This can only be achieved by encrypting part of the data that are considered to be extremely sensitive and leave the remaining part. The motive behind this approach is to reduce to the barest minimum the cost and time of applying the encryption on the entire data [46]. This approach however requires verification as healthcare providers will not rely on theoretical solution without any empirical evaluation. In conclusion, it is very clear that e-Health system is undoubtedly a good IT initiative to enhance the usage of health all over the world. The issue of security and privacy is however very sacrosanct. To ensure efficient utilization of this innovation; security and privacy issues should be adequately taken care of.

Table 2 provides the summary of reviewed work carried out in this research.

Table 2
Summary of reviewed articles.

Author	Year	Approach	Strength	Weakness
Shin et. al.	2014	Extended Role Based Access Control (xRBAC)	Seamless information sharing	Not suitable for a distributed environment and it is not scalable
Simplicio et al.	2015	Transport Layer Security/Secure Sockets Layer (TLS/SSL)	It prevents alien from unauthorized access	It is platform dependent and not scalable
Barua et al	2011	Attribute Based Encryption (ABE)	privileges were mapped into various roles with ABE access structures	storage of health information is located in a centralized server
Guo et al.	2012	Decentralization of authorization server	Users are permitted to access based on their privileges without disclosing their attributes and identities	There is no room for collaborative sharing of medical data across different domains. It lacks interoperability
Gajanayake et al.	2016	Discretionary Access Control (DAC), Mandatory Access Control (MAC) and Role Based Access Control (RBAC)	Patients do dictate and set access privileges	The model cannot be used in a distributed environment
Barua et. al	2011	Proxy Re-encryption	The schema has a good and excellent performance	The schema doesn't give room for scalability and flexibility
Kumar et al.	2013	Attribute Based Encryption (ABE)	It handles key management complexity	It is neither scalable nor flexible.
Zhu et al.	2014	Re-encryption and Attribute Based Encryption (ABE) with proxy encryption that is Rivest Shamir and Adleman (RSA) enabled	Healthcare practitioner can easily be prevented from getting the read keys without approval from both end	Limited number of users. It is not scalable.
Sunagar and Biradar	2014	Advanced Encryption Standard (AES) algorithm	Reliable and secured in a cloud-based environment	It is operating system dependent
Liu et al.	2015	Role-Based Access Control (RBAC) with Hierarchical Identity-Based Encryption (HIBE) schema	Efficient with reliable	It doesn't provide reliable and accurate access control requirements
Bahtiyar and Cag Jayan	2014	A trust-based assessment security model	Better results in terms of trust computation when compared to the various existing trust models	It contains mathematical variables whose values cannot clearly be evaluated
Shin et al.	2014	An enhanced Role Based Access Control model	Efficient with reliable	Unusable in a collaborative environment
Li and Hoang	2009	A multi-agent system	It initiates various requests to interact in a flexible manner	Lacks security information aggregation
Fan et al.	2014	Single Point of Contact (SPoC)	It facilitates integration and deployment of reliable e-Health services	It is not flexible and dynamic enough for a very large number of users
Bhartiya et al.	2015	Hierarchy Similarity Analyzer (HSA).	It evaluates and assigns a Security_Level (SL) to users sharing data	There is no absolute guarantee to security and privacy in any federated agencies.
Rezaeibagha and Mu	2016	Hybrid clouds as well as access control policy transformation and cryptographic building blocks	Different users of EHR with different access privileges and permissions	It doesn't give room for scalability
Morchon and Wehrle	2010	RBAC security model	It is easy and convenient for deployment of policies on various sensor networks	It lacks detection mechanism for illegal and unauthorised access in case of any emergency situation
Amini et al.	2011	MiniSec, RC4-based as well as various ciphers algorithms such as RC4 and Advanced Encryption Standard (AES)	It is very efficient and reliable for achieving access confidentiality and integrity in Electronic Health	The conclusion drawn on the few algorithms is not reliable enough
Wang et al.	2017	Identity Based Encryption (IBE) and new Identity Based Proxy Re-encryption (IBPRE) schemes	Newly developed IBPRE is better and efficient for re-encryption	The performance of the model is not reliable and efficient
Karakış et al.	2015	Steganographic approach	The message is protected to ward off any form of attack through lossless compression and symmetric encryption algorithms	It could not handle noise cancellation and data reduction.
Sahi et al.	2016	Privacy-Preserving approach and the Security-Preserving approach and a disaster recovery	The former approach is a robust mechanism for achieving both the privacy and integrity of medical information while the latter approach could only be used for reliable	Inability of the mechanism to function efficiently if any of the approaches fails
Peleg et al.	2008	A Situation-Based Access Control (SitBAC) model	Model is considered to be generic which can also be used and adapted in other domains	It could not be able to factor in all the likely stakeholders with diverse goals
Rubio et al.	2013	Modified the security features in the Standard communications protocol	It is very effective to authorise and authenticate users to protect the privacy of sensitive electronic health information	It is very technical and difficult to implement
Martínez et. al.	2013	Anonymization (Statistical Disclosure Control (SDC))	It provides accurate and reliable results of SDC approaches	This framework cannot also work for large dataset. Also, the framework generates a significant delay during the anonymization process

(*continued on next page*)

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001

Table 2 (*continued*)

Author	Year	Approach	Strength	Weakness
Kim et al.	2014	Delay-free anonymization	This approach doesn't generate time delay since stream of data are anonymized immediately with fake values	It doesn't investigate the timing for the late validation
Barua et al.	2011	Role Based Access Control (RBAC)	Permits information requester on electronic health platform to have access privileges based on their assigned roles and their corresponding attributes	Other forms of Role Based Access Control (RBAC) were not considered
Kester et al.	2015	Watermarking and encryption	It efficiently secures and authenticates medical images.	The framework is limited to images alone
Yu et al. 2010	2102	Lazy re-encryption, proxy re-encryption and attribute-based encryption (ABE)	A heavy computation overhead and data management problems were resolved	It is not fully scalable and did not provide efficient and reliable data confidentiality and integrity
Löhr et al.	2010	Access control architecture	It provides a combination of users and network security	Inability to remember a PIN to access
Hupperich et al.	2012	Modern cryptographic security approach	The access is time independent and not place-bound	There is complexity in key management
Kahani et al.	2016	DUKPT and a two-stage combination of key encryption was adopted	It tolerates a good number of simultaneous authorization requests with excellent response time	It has limited number of entities: users U, a cloud server CS, a service provider SP, and an authentication and access control manager AAM
Li et al.	2010	Attribute Based Encryption (ABE) security model	It reduces the complexity that may arise due to key distribution	It fails to support more expressive owner-defined access control policies
Li et al.	2012	Attribute Based Encryption (ABE)	It reduces the key management complexity for the owners and users	The approach is not scalable

5. Future directions on security of e-Health

It has been established that remarkable progress has been made regarding security and privacy in e-Health in cloud-based environment. It is also very significant to suggest few approaches to ensure and enforce security measures in e-Health system. Doing this is aimed at maintaining and enhancing efficiency of the security and privacy initiatives. Hence the followings strategies are proposed as the new security solutions:

a. We are of the opinion that auditing will assist at ensuring security and privacy in e-Health. It will assist in locating and identifying any form of misconduct that could affect e-Health solution. Hence, auditing should be considered as a new research direction for e-Health solution;
b. From the available and reviewed literatures, we discovered that most e-Health solutions use encryption scheme to ensure and achieve security and privacy. We are of the opinion that encrypting the parts or sections which reveal the information about the e-Health user will be excellent as the remaining part of the data is left unencrypted. Doing this will not in any way leads to insecurity of data;
c. Most solutions utilized RBAC model to ensure security and privacy in e-Health. We are of the view that Attribute Based Access Control (ABAC) model should be used to ensure excellent scalability and flexibility for authentications and authorizations;
d. Attribute Based Encryption (ABE) is known to be excellent in ensuring privacy in e-Health, but excessive computations while decrypting data is imminent and affecting its performance because of bi-linear pairing operations. We are also of the opinion that finding solution to this bi-linear operations will enhance the efficiency of ABE. Searching for the solution is considered a good research area for e-Health;
e. General enforcement of privacy requirements should be adopted. Most of the solutions focused mainly on the security and privacy of patients. Privacy of all parties involved in electronic health systems should be incorporated. Getting the required security and privacy violations will be a good research area;
f. Based on the literatures reviewed, it was noted that some solutions adopted RBAC, MAC, and DAC. We are of opinion that these models will perform better if they are hybridized to form a single model for ensuring security and privacy in e-Health. This is a very good and interesting area of research which researchers could focus on in e-Health.

With all the available literatures, particularly those reviewed above, it is very clear that nearly all the existing models are suffering from one challenge or the other. Majorly, some of the current models are having challenges of scalability, interoperability, flexibility, compatibility, improper model evaluation, inability to implement in a distributed environment such as cloud computing, key management, cost and time complexities among others. In order to have a dependable framework security and privacy for e-Health solutions, all these security challenges should be properly taken care of.

Fig. 4 shows the proposed architecture for privacy in e-health. The performance of the architecture is expected to provide high degree of reliability, dependability and efficiency in protecting patient information after its full implementation. Specifically, the architecture is expected to give room for flexible authentication to e-Health information on patient. Also, it is expected to give absolute control of information by the major stakeholders as well as providing mechanism for emergency and complex situation whenever the need arises.

6. Features of proposed system

The motive behind this architecture is to build a secured, dynamic and dependable framework for E-Health. The architectural framework is to be absolutely controlled by the patient who is considered a major stakeholder in E-health system.

A patient will to obtain his full access through authentication into a designated medical institution (hospital). At this point, he chooses who his medical personnel is. A medical officer (MO) will have access to patient's information that is available in the cloud by using Access Control List (ACL) security model [36]. The type of activity to carry out on a patient will be regulated by Mandatory Access Control (MAC) [36] security model.

Mandatory Access Control (MAC) is a type of security model that permits operating system to coerce the ability of an initiator of an action to carry out some operations on a target or an object. An object is regarded as constructs such as files, IO devices or

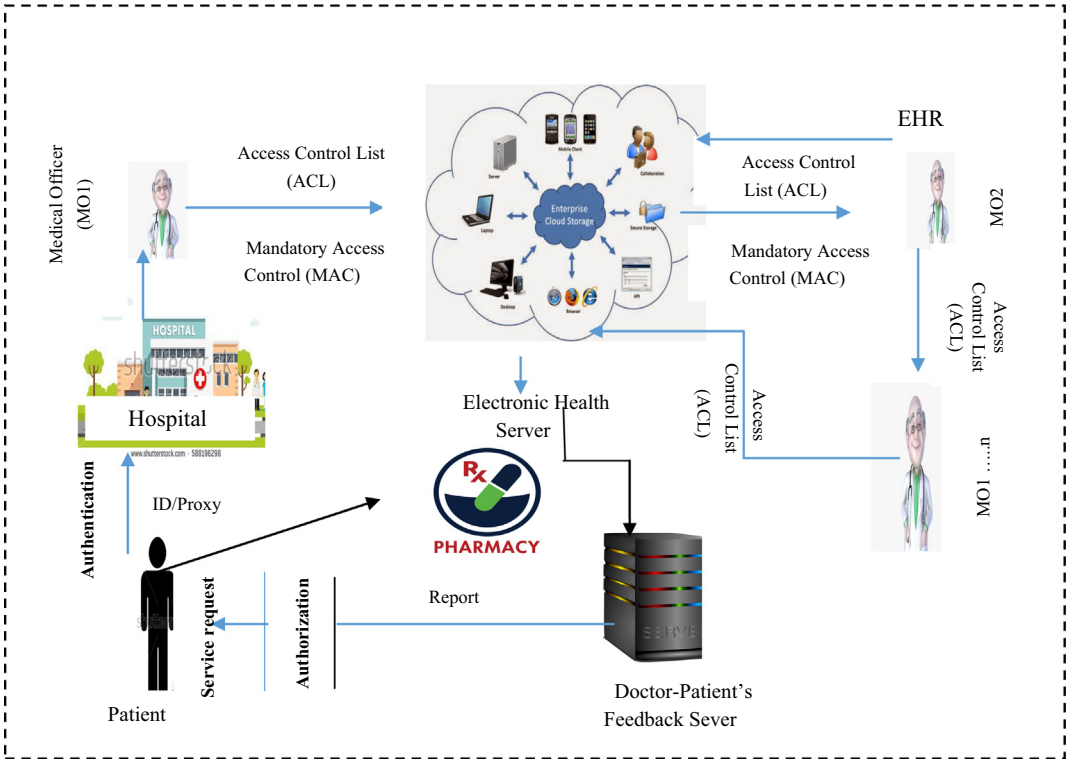

Fig. 4. Proposed Security Architecture for Electronic Health.

memory segments while a subject could be regarded as a thread or a process.

Access Control List (ACL) on the other hand is a security model that provides a set of permission attached to objects. It indicates who among the users or processes should be granted access privilege to objects. It also specifies what operations are permitted on given objects [36].

The architecture permits proxy to give room for quick attendance to any patience in case of an emergency.

7. Conclusion

It is very vital to implement e-Health solution in any country of the world to enhance excellent healthcare delivery system. To maximally enjoy the services of e-Health, it is very important and fundamental to put in place the required security and privacy mechanisms to prevent any form of security breach and vulnerability. From the foregoing, we have been able to review literatures on security and privacy in e-Health and also identified lapses in the existing solutions. In order to have efficient e-Health solution, it is important we incorporate some of the suggested solutions proposed in any model for e-Health solution. The government and policy makers in all countries of the world should develop a comprehensive e-Health document framework to motivate and enable its acceptance. Governments should also develop research institutes where security experts will come together to brainstorm on how to develop e-Health solution that would be secured enough from any form of vulnerability. Finally, there should be a detailed privacy regulations on the services and practices of e-Health so that patients can feel highly protected while disclosing their health-related information.

References

[1] Zhang R, Liu L. Security models and requirements for healthcare application clouds. In: 3rd IEEE International Conference on Cloud Computing (CLOUD), Miami, FL, USA, USA, pp. 268–275 (2010).

[2] Abbas A, Bilal K, Zhang L, Khan S. A cloud based health insurance plan recommendation system. Future Gen Comput Syst 2015;43:99–109.

[3] Abbas, A, Khan, M, Ali, M, Khan, S, Yang, L. A cloud based framework for identification of influential health experts from Twitter. In: Proceedings of the 15th International Conference on Scalable Computing and Communications (ScalCom) (2015), Beijing, China, pp.831-838 (2015)

[4] Li M, Yu S, Zheng Y, Ren K. Scalable and secure sharing of personal health records in cloud computing using attribute-based encryption. IEEE Trans Parallel Distrib Syst 2012;1–14.

[5] Eysenbach G. What is e-health? J Med Internet Res 2001;3(2).

[6] Subashini S, Kavitha V. Review: a survey on security issues in service delivery models of cloud computing. J Netw Comput Appl 2011;34(1):1–11.

[7] Abbas, A, Khan, SU, e-Health Cloud: Privacy Concerns and Mitigation Strategies. In: Medical Data Privacy Handbook. Springer International Publishing, Switzerland (2015) pp. 389–421.

[8] Zissis D, Lekkas D. Addressing cloud computing security issues. Future Gener Comput Syst 2012;28(3):583–92.

[9] VMware: Your cloud in healthcare. In: http://www.vmware.com/files/pdf/VMware-Your-Cloudin-Healthcare-Industry-Brief.pdf. (Accessed 2015) –>

[10] Skorin-Kapov L, Matijasevic M. Analysis of QoS requirements for e-health services and mapping to evolved packet system QoS classes. Int J Telemed Appl 2010;1–19.

[11] Pussewalage H, Oleshchuk V. Privacy preserving mechanisms for enforcing security and privacy requirements in e-health solutions. Int J Inf Manage 2016;1161–1173.

[12] Shin M, Jeon H, Ju Y, Lee B, Jeong S. Constructing RBAC based security model in u-healthcare service platform. Sci World J 2014;1–13.

[13] Simplicio M, Iwaya L, Barros B, Carvalho T, Naslund M. SecourHealth: a delay-tolerant security framework for mobile health data collection. IEEE J Biomed Health Inform 2015;19(2):761–72.

[14] Barua, M, Lu, R, Liang, X, Shen, X, PEACE: An Efficient and Secure Patient-centric Access Control Scheme for eHealth Care System. In: The First International Workshop on Security in Computers, Networking and Communications, Shanghai, China, 2011, pp. 970–975.

[15] Guo, L, Zhang, C, Sun, J, Fang, Y. PAAS: A Privacy-Preserving Attribute-based Authentication System for eHealth Networks. In: 2012 32nd IEEE International Conference on Distributed Computing Systems, Macau, China, 2012, pp. 224–233.

[16] Azeez, N, Ademolu, O. CyberProtector: Identifying Compromised URLs in Electronic Mails with Bayesian Classification. In: International Conference Computational Science and Computational Intelligence (CSCI), Las Vegas, NV, USA, 2016, pp. 959–965.

[17] Gajanayake R, Iannella R, Sahama T. Privacy oriented access control for electronic health records. e-J Health Inf 2014;8(2):175–86.

[18] Kumar M, Fathima M, Mahendran M. Personal health data storage protection on cloud using MA-ABE. Int J Comput Appl 2013;75(8):11–6.

[19] Zhu, H, Huang, R, Liu, X, Li, H, SPEMR: A new secure personal electronic medical record scheme with privilege separation. In: 2014 IEEE International

Conference on Communications Workshops (ICC), Sydney, NSW, Australia, 2014, pp. 700–705.
[20] Sunagar V, Biradar C. Securing public health records in cloud computing patient centric and fine grained data access control in multi owner settings. Int J Sci Appl Inf Technol (IJSAIT) 2014;3(4):18–21.
[21] Liu, W, Liu, X, Liu, J, Wu, Q, Zhang, J, Li, Y. Auditing and Revocation Enabled Role-Based Access Control over Outsourced Private EHRs. In: 2015 IEEE 17th International Conference on High Performance Computing and Communications (HPCC), New York, NY, USA, 2015, pp. 336–341.
[22] Bahtiyar S, Çağlayan M. Trust assessment of security for e-health systems. Electron. Commer. Res. Appl. 2014;164–177.
[23] Li, W, Hoang, D., A new security scheme for e-health system. In: International Symposium on Collaborative Technologies and Systems, 2009. CTS '09., Baltimore, MD, USA, 2009, pp. 361–366.
[24] Fan, L, Lo, O, Buchanan, W, Ekonomou, E, Sharif, T, Sheridan, C., SPoC: Protecting Patient Privacy for e-Health Services in the Cloud., 2014, pp. 1–6.
[25] Bhartiya, S, Mehrotra, D, Girdhar, A.: Proposing hierarchy-similarity based access control framework: A multilevel Electronic Health Record data sharing approach for interoperable environment. Journal of King Saud University – Computer and Information Sciences, 1-15 (August 2015)
[26] Rezaeibagha F, Mu Y. Distributed clinical data sharing via dynamic access-control policy transformation. Int J Med Inf 2016:25–31.
[27] Garcia-Morchon, O, Wehrle, K. Efficient and context-aware access control for pervasive medical sensor networks. In: 2010 8th IEEE International Conference on Pervasive Computing and Communications Workshops (PERCOM Workshops), Mannheim, Germany, 2010, pp. 322–327.
[28] Amini, S, Verhoeven, R, Lukkien, J, Chen, S, Toward a security model for a body sensor platform. In: 2011 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 2011, pp. 143–144.
[29] Wang X, Ma J, Xhafa F, Zhang M, Luo X. Cost-effective secure E-health cloud system using identity based cryptographic techniques. Future Gener Comput Syst 2017;67:242–54.
[30] Karakış R, Güler I, Çapraz I, Bilir E. A novel fuzzy logic-based image steganography method to ensure medical data security. Comput Biol Med 2015;172–183.
[31] Sahi A, Lai D, Li Y. Security and privacy preserving approaches in the eHealth clouds with disaster recovery plan. Comput Biol Med 2016;78:1–8.
[32] Peleg M, Beimel D, Dori D, Denekamp Y. Situation-based access control: privacy management via modeling of patient data access scenarios. J Biomed Inform 2008;41:1028–40.
[33] Rubio O, Alesanco A, García J. A robust and simple security extension for the medical standard SCP-ECG. J Biomed Inf 2013;142–151.
[34] Martínez S, Sánchez D, Valls A. A semantic framework to protect the privacy of electronic health records with non-numerical attributes. J Biomed Inf 2013:294–303.
[35] Kim S, Sung M, Chung Y. A framework to preserve the privacy of electronic health data streams. J Biomed Inf 2014:95–106.
[36] Barua M, Liang X, Lu R, Shen X. ESPAC: enabling security and patient-centric access control for e-Health in cloud computing. Int J Security Netw 2011:67–76.
[37] Kester, Q, Nana, L, Pascu, A, Gire, S, Eghan, J, Quaynor, N. A Security Technique for Authentication and Security of Medical Images in Health Information Systems. In: 2015 15th International Conference on Computational Science and Its Applications, Banff, AB, Canada, 2015, pp. 8–13.
[38] Wu, R, Ahn, G, Hu, H. Secure Sharing of Electronic Health Records in Clouds. In: 8th International Conference Conference on Collaborative Computing: Networking, Applications and Worksharing, Collaboratecom 2012 Pittsburgh, PA, United States, October 14-17, 2012, Pittsburgh, PA, United States, 2012, pp. 711–718.
[39] Yu, S, Wang, C, Ren, K, Lou, W. Achieving Secure, Scalable, and Fine-grained Data Access Control in Cloud Computing. In: 2010 Proceedings IEEE, INFOCOM, San Diego, CA, USA, 2010 pp. 1–9.
[40] Löhr, H, Sadeghi, A, Winandy, M. Securing the E-Health Cloud. In: 1st ACM International Health Informatics Symposium (IHI 2010)., Arlington, Virginia, USA, 2010, pp. 1–10.
[41] Hupperich, T, Löhr, H, Sadeghi, A, Winandy, M. Flexible Patient-Controlled Security for Electronic Health Records. In: 2nd ACM SIGHIT International Health Informatics Symposium (IHI 2012)., Miami, Florida, USA, 2012, pp. 1–5.
[42] Kahani, N, Elgazzar, K, Cordy, K. Authentication and Access Control in e-Health Systems in the Cloud. In: IEEE International Conference on High Performance and Smart Computing (HPSC), Big Data Security on Cloud (BigDataSecurity), New York, NY, USA, 2016, pp. 13–23.
[43] Li, M, Yu, S, Ren, K, Lou, W. Securing Personal Health Records in Cloud Computing: Patient-Centric and Fine-Grained Data Access Control in Multi-owner Settings. In: International Conference on Security and Privacy in Communication Systems, Singapore, Singapore, 2010, pp. 89–106.
[44] Azeez N, Venter I. Towards ensuring scalability, interoperability and efficient access control in a multi-domain grid-based environment. SAIEE Afr Res J 2013:54–68.
[45] Ayofe AN, Adebayo SB, Ajetola AR, Abdulwahab AF. A framework for computer aided investigation of ATM fraud in Nigeria. Int J Soft Comput 2010;5(3):78–82.
[46] Azeez NA, Olayinka AF, Fasina EP, Venter IM. Evaluation of a flexible column-based access control security model for medical-based information. J Comput Sci Its Appl 2015;22(1):14–25.
[47] Azeez NA, Babatope AB. AANtID: an alternative approach to network intrusion detection. J Comput Sci Appl Int J Nigeria Comput Soc 2016:129–43.
[48] Azeez NA, Iliyas HD. Implementation of a 4-tier cloud-based architecture for collaborative health care delivery. Nigerian J Technol Dev 2016;13(1):17–25.
[49] Azeez NA, Iyamu T, Venter IM. Grid security loopholes with proposed countermeasures. In: Gelenbe E, Lent R, Sakellari G, editors. 26th International Symposium on Computer and Information Sciences. London: Springer; 2011. p. 411–8.
[50] Azeez NA, Lasisi AA. Empirical and statistical evaluation of the effectiveness of four lossless data compression algorithms. Nigerian J Technol Dev 2016;13(2):64–73.
[51] Nureni AA, Irwin B. Cyber security: challenges and the way forward. Comput Sci Telecommun 2010;29:56–69.
[52] Gai K, Qiu M, Sun X. A survey on FinTech. J Netw Comput Appl 2018;103(2018):262–73.
[53] Gai K, Qiu M. Blend arithmetic operations on tensor-based fully homomorphic encryption over real numbers. IEEE Trans Ind Inf 2018;14(8):3590–8.
[54] Gai K, Choo KR, Qiu M, Zhu L. Privacy-preserving content-oriented wireless communication in internet-of-things. IEEE Int Things J 2018;5(4):3059–67.
[55] Gai K, Qiu M, Xiong Z, Liu M. Privacy-preserving multi-channel communication in Edge-of-Things. Future Gener Comput Syst 2018;85:190–200.
[56] Liu X, Lu R, Ma J, Chen L, Qin B. Privacy-Preserving Patient-Centric Clinical Decision Support System on Naive Bayesian Classification. IEEE J Biomed Health Inf 2016;20(2).
[57] Liu, X, Deng, RH, Choo, KR, Yang, Y. Privacy-Preserving Outsourced Clinical Decision Support System in the Cloud. DOI 10.1109/TSC.2017.2773604, IEEE Transactions on Services Computing.

Please cite this article as: N. A. Azeez and C. V. der Vyver, Security and privacy issues in e-health cloud-based system: A comprehensive content analysis, Egyptian Informatics Journal, https://doi.org/10.1016/j.eij.2018.12.001